

1. COURSE CONTENT

School/Programme	EEE / Master of Science (CCA)
Course Code	EE6221
Course Title	Robotics and Intelligent Sensors
Pre-requisites	Linear Control Theory and Linear Algebra
Contact Hours	39

Course Aims

This course introduces advanced topics in robotics and intelligent sensing techniques. The objectives of the course are to provide an understanding of robotics and intelligent sensors. Students will be exposed to a broad range of topics in robotics and intelligent sensors, with emphasis on manipulator kinematics, trajectory planning, nonlinear control techniques, hybrid position and force control, mobile robot kinematics, robot sensors basics, vision sensors, estimation using vision sensors, sensor integration and fusion, and visual servo control.

Intended Learning Outcomes (ILO)

On completion of this course, the student will be able to:

1. model kinematics of robot manipulators.
2. solve an inverse kinematics problem and plan a robot task.
3. design and analyze nonlinear robot controllers and hybrid position and force controllers using appropriate methods.
4. model kinematics of mobile robots.
5. understand classification, characteristics and features of main internal and external sensors.
6. explain the modelling, calibration, and application scenarios of vision sensors.
7. explain essential matrix and homography matrix techniques; analyse structure from motion and pose reconstruction using essential matrix and homography matrix techniques.
8. design estimators based on Kalman filter techniques for sensor fusion; design and analyze visual servo controllers.

Course Content

Overview and Kinematics (11 hours)

Classification of robots. Specifications. Homogeneous transformation. Denavit-Hartenberg coordinate transforms. Forward kinematics. Inverse kinematics. Trajectory Planning. Manipulator Jacobian. Resolved Motion Rate Control.

Advanced Control Techniques (5 hours)

Nonlinear control. Computed torque control. Performance specifications. Robot Dynamic Model. Force Control. Hybrid position and force control.

Mobile Robot (5 hours)

Locomotion and Wheeled Mobile Robots. Wheels and Kinematic Constraints. Kinematic modelling.

Sensor Basics and Systems (6 hours)

Classes of sensors. Internal state sensors. External state sensors. Vision systems. Imaging components. Image representation and camera model.

Sensing and Sensor Fusion (6 hours)

Structure from Motion. Pose reconstruction. Essential matrix. Homography matrix. Kalman filter. Sensor integration and fusion.

Visual Servo Control (6 hours)

Visual servoing introduction. Position-based visual servo control. Image-based visual servo control.

Assessment (includes both continuous and summative assessment)

Component	Weighting
Final Examination	60%
Continuous Assessment 1 (CA1): Quiz 1	15%
Continuous Assessment 2 (CA2): Quiz 2	15%
Continuous Assessment 3 (CA3): Assignment	10%

Reading and References

1. Schilling R. J., Fundamentals of Robotics: Analysis and Control, Prentice Hall, 1990.
2. Craig J. J., Introduction to Robotics, Mechanics and Control, PEARSON. Prentice. Hall. 2005.
3. Siegwart R. and Nourbakhsh I. R., Introduction to Autonomous Mobile Robots, The MIT Press, 2004.
4. Stadler W., Analytical Robotics and Mechatronics, McGraw-Hill, 1995.
5. Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Cambridge University Press, 2004.
6. Olivier Faugeras, Quang-Tuan Luong, and Theodore Papadopoulos, The Geometry of Multiple Images, MIT Press, 2001.
7. Yi Ma, Stefano Soatto, Jana Kosecka, and S. Shankar Sastry, An Invitation to 3-D Vision: From Images to Geometric Models, Springer-Verlag New York, 2004.

Course Instructors

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